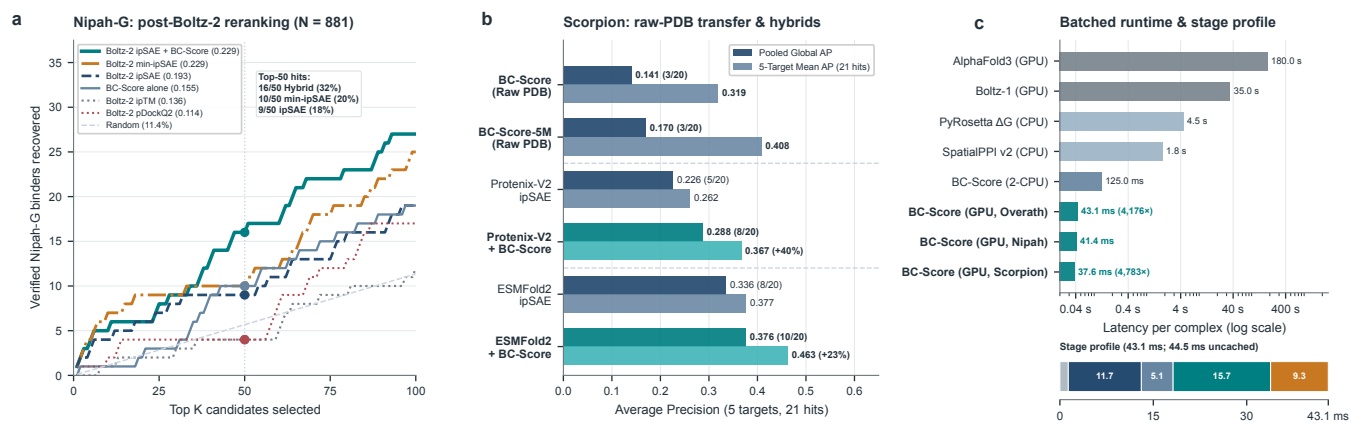

a
Nipah-G: post-Boltz-2 reranking (N = 881)
Boltz-2 ipSAE + BC-Score (0.229)
Boltz-2 min-ipSAE (0.229)
Boltz-2 ipSAE (0.193)
BC-Score alone (0.155)
Boltz-2 ipTM (0.136)
Boltz-2 pDockQ2 (0.114)
Random (11.4%)
Top-50 hits:
16/50 Hybrid (32%)
10/50 min-ipSAE (20%)
9/50 ipSAE (18%)
Verified Nipah-G binders recovered
Top K candidates selected
b
Scorpion: raw-PDB transfer & hybrids
Pooled Global AP
5-Target Mean AP (21 hits)
BC-Score (Raw PDB)
0.141 (3/20)
0.319
BC-Score-5M (Raw PDB)
0.170 (3/20)
0.408
Protenix-V2 ipSAE
0.226 (5/20)
0.262
Protenix-V2 + BC-Score
0.288 (8/20)
0.367 (+40%)
ESMFold2 ipSAE
0.336 (8/20)
0.377
ESMFold2 + BC-Score
0.376 (10/20)
0.463 (+23%)
Average Precision (5 targets, 21 hits)
c
Batched runtime & stage profile
AlphaFold3 (GPU)
180.0 s
Boltz-1 (GPU)
35.0 s
PyRosetta ΔG (CPU)
4.5 s
SpatialPPI v2 (CPU)
1.8 s
BC-Score (2-CPU)
125.0 ms
BC-Score (GPU, Overath)
43.1 ms (4,176×)
BC-Score (GPU, Nipah)
41.4 ms
BC-Score (GPU, Scorpion)
37.6 ms (4,783×)
Latency per complex (log scale)
Stage profile (43.1 ms; 44.5 ms uncached)
11.7
5.1
15.7
9.3
43.1 ms